

# Knowledge Management in Globally Distributed Agile Projects—Lesson Learned

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**Abstract**—Knowledge management (KM) is essential for success in any software project, but especially in global software development where team members are separated by time and space. Software organizations are managing knowledge in various ways to increase transparency and improve software team performance. One way to classify these strategies is proposed by Earl who defined seven knowledge management schools. The objective of this research is to study knowledge creation and sharing practices in a number of distributed agile projects, map these practices to the knowledge management strategies and determine which strategies are most common, which are applied only locally and which are applied globally. This is done by conducting a series of semi-structured qualitative interviews over a period of time span during May, 2012-June, 2013. Our results suggest that knowledge sharing across remote locations in distributed agile projects heavily relies on knowledge codification, i.e. technocratic KM strategies, even when the same knowledge is shared tacitly within the same location, i.e. through behavioral KM strategies.

**Index Terms**—Knowledge management, knowledge sharing, distributed, global software engineering, agile, shared understanding.

## I. INTRODUCTION

Knowledge management (KM) is an essential area in software project management, since software development is a very knowledge-intensive job. KM is also often considered a software process improvement initiative [1,2]. One of the biggest challenges in software organizations is to ensure availability of the right knowledge at the right time due to knowledge fragmentation, overload and de-contextualization [3]. Agile software projects address these challenges by emphasizing social interaction and conversations [4]. Knowledge is managed primarily through socialization strategies [5]. While face-to-face interaction facilitates effective knowledge sharing between team members, this is difficult to achieve in distributed projects, in which team members are dispersed across multiple locations and hardly ever meet in person. Distributed development is often associated with rigorous standardization and thus knowledge management strategies used are primarily characterized by knowledge codification or externalization [5].

Nowadays, managers often strive to apply agile practices in distributed projects, and therefore the question of how to organize knowledge management in such environments emerges. Distributed agile projects are in need of finding ways to ensure process agility and at the same time addressing

the challenges of distribution. In this paper we explore seven distributed agile projects and summarize our lessons learned.

The rest of the paper is organized as follows. In Section II we outline related research that is used as the background for our investigation. Section III contains the details regarding our research methodology. The results from studying seven distributed agile projects are presented in Section IV followed by the discussion in section V and conclusions in Section VI.

## II. BACKGROUND

Earl [6] proposed a classification of knowledge management practices which we use as the key basis for our empirical investigation. This classification suggests to distinguish knowledge management practices into three broad categories: technocratic, economic, and behavioral. These are further divided into seven schools: Systems, Cartographic and Engineering (*Technocratic*), Commercial (*Economic*) and Organizational, Spatial and Strategies (*Behavioral*). These are described as follows.

**Systems:** This school's philosophy is to codify knowledge with the help of technology. Organizations intention is to use knowledge repositories for storing and sharing knowledge. These knowledge repositories are usually used to store domain specific information that can be used for decision-making purposes. The codification of knowledge can be compared with externalization of knowledge by Nonaka [5]. It is easy to realize the benefits of knowledge bases. That is why *Systems school* is the most researched school [7]. These knowledge bases become richer and more useful overtime.

**Cartographic:** This school strategically places a knowledgeable person to be easy accessible for others in an organization to consult or exchange knowledge. This is sometimes achieved by searchable knowledge repositories or "yellow-pages". Ericsson uses a similar idea by introducing knowledge brokers [1]. Knowledge brokers are knowledgeable and experienced software engineers who will communicate with other developers with information or will listen to them.

**Engineering:** This KM school focuses on business process reengineering [7]. The two main ideas of this school are twofold; *Firstly*, business process performance can be enhanced by providing knowledge to the relevant personnel. *Secondly*, management processes are more complex than business processes, hence providing contextual knowledge can be

TABLE I: Overview of Earl's Framework [6]

| Categories   | Schools        | Focus      | Unite       | Aim                    | Philosophy        |
|--------------|----------------|------------|-------------|------------------------|-------------------|
| Technocratic | Systems        | Technology | Domain      | Knowledge Bases        | Codification      |
|              | Cartographic   | Maps       | Enterprise  | Knowledge Directories  | Connectivity      |
|              | Engineering    | Process    | Activity    | Knowledge Flows        | Capability        |
| Economic     | Commercial     | Income     | Know-how    | Knowledge Assets       | Commercialization |
| Behavioral   | Organizational | Networks   | Communities | Knowledge Pooling      | Collaboration     |
|              | Spatial        | Space      | Place       | Knowledge Exchange     | Connectivity      |
|              | Strategic      | Mindset    | Business    | Knowledge Capabilities | Consciousness     |

beneficial. Software process improvements like CMMI can be regarded as stimuli for knowledge flow throughout the organization. Small and Medium size organizations can adopt tailored agile practices and be benefited.

**Commercial:** This classification of knowledge management focuses on economic value of knowledge as an asset. Knowledge in an organization becomes obviously sellable when it is more domain specific or expert level. Usually organizational knowledge or “*intellectual property*” is translated into patents, trademarks, copyright etc.

**Organization:** This KM school focuses on the organizational structure. These structures are often referred as “knowledge communities” [7]. This is a networking approach for people to communicate and share knowledge. The main aim is to collaborate and create knowledge for the organization.

**Strategic:** In this school of knowledge management organizations identify strategies that add competitive advantages. It is related to developing conceptual models knowledge management for organizations.

**Spatial:** This KM school focuses on using organizations space for knowledge sharing. Organizations use different office settings to encourage people to communicate. For example in the case of software organizations agile methodologies promote the use of boards, charts or other tools to create spatial knowledge. Sometimes even common spaces like conference rooms, coffee rooms or place for refreshment activities also serve as places where the knowledge can be shared.

In the light of our research focus on distributed agile projects, we can say that naturally agile projects would be associated with behavioral knowledge management strategies, while distributed projects with technocratic strategies. However, current research has advised that software organizations should carefully consider using several strategies [8]. In fact, some global software organizations find ways to adopt behavioral strategies in distributed settings in new ways that are not used in collocated projects through e.g. rotating team members to different locations to work with remote members for a specific period of time [9]. Through a case study Razzak et al. [10] identified the use of spatial knowledge in a distributed agile project. The studied team used different techniques to create spatial knowledge in collocation but no spatial knowledge was created globally with the remote team members. The authors also concluded that, practitioners are more focused on systems and organizational schools rather than the spatial school due to a lack of inexpensive software, hardware and processes.

In our previous research [11] we have also reported knowledge management strategies in distributed agile projects. The current research focuses on exploring knowledge sharing strategies in more detail. We were thus driven by the following research question:

- Which knowledge management strategies are used by distributed agile teams and how?

### III. RESEARCH METHODOLOGY

Knowledge sharing strategies were investigated in an interview-based study of a number of distributed agile projects. The first author performed a series of semi-structured interviews in six different organizations during May 2012 – June 2013.

#### A. Sampling of the Interviewees

The selection of interviewees in this study followed convenience sampling [12], which means that the contacts of the authors were utilized to find the case projects. We targeted software companies that have no less than 2 years of experience with distributed agile development. We opted to approach potential interviewees with a sound track record and experience, which was screened using LinkedIn profiles. The potential interviewees received an invitation via email. We sought interviewees currently involved in agile distributed projects. Finally, seventeen interviewees from six different companies committed their participation in our study (see also Table II).

The selected companies are involved in software product development, have different organizational settings and structure and are located in different countries. The participants of this research were project managers, team leaders, software architects, line managers, senior software developers, system developers and Scrum masters in different countries involved in distributed agile projects, located in different countries i.e. Sweden, Norway, Germany, Ukraine, China, India, Bangladesh, USA, and Latvia. To get the rounded perspective of this research phenomenon we included different roles from the agile team.

#### B. Data Collection

Due to the exploratory nature of this study we used semi-structured interviews. According to Robson [13], an in-depth semi-structured interview is helpful in finding out what is happening and seeking new insights. Seventeen semi-structured interviews were conducted from seven teams in

**TABLE II:** Overview of distributed Agile projects

| <i>Projects</i> | <i>Project Distribution</i> | <i>Team Size</i> | <i>Team Types</i> | <i>Agile Position/Roles</i> | <i>No. of Interviewees</i> |
|-----------------|-----------------------------|------------------|-------------------|-----------------------------|----------------------------|
| <b>Alpha</b>    | Sweden-Germany              | 6-7*             | Dispersed         | Team Leader                 | 1                          |
|                 |                             |                  |                   | Developer                   | 1                          |
| <b>Beta</b>     | Norway-Bangladesh           | 5-6*             | Dispersed         | Project Manager             | 1                          |
|                 |                             |                  |                   | Developer                   | 1                          |
| <b>Gamma</b>    | USA-Bangladesh              | 12-16**          | Distributed       | Head of Engineering         | 1                          |
|                 |                             |                  |                   | Senior Developer            | 1                          |
|                 |                             |                  |                   | Developer                   | 1                          |
| <b>Delta</b>    | Sweden-Bangladesh           | 16-18**          | Dispersed         | Software Architect          | 1                          |
|                 |                             |                  |                   | Developer                   | 1                          |
| <b>Epsilon</b>  | Latvia- Ukraine             | 11-15**          | Distributed       | Project Manager             | 1                          |
|                 |                             |                  |                   | Developer                   | 1                          |
| <b>Zeta</b>     | Sweden-China                | 26-35***         | Distributed       | Line Manager                | 1                          |
|                 |                             |                  |                   | Software Developers         | 2                          |
|                 |                             |                  |                   | System Developer            | 1                          |
| <b>Eta</b>      | Sweden-India                | 45-55***         | Hybrid            | System Developer            | 1                          |
|                 |                             |                  |                   | Scrum Master                | 1                          |

*In Table II \*, \*\*, \*\*\* indicates Small, Medium and Large teams respectively*

order to identify how practitioners are creating, storing and sharing knowledge related to software development among geographically distributed agile teams. Before the interviews started, we discussed the overall goals. The interview questions were descriptive. Follow up questions were asked based on the discussion. These concerned shared knowledge creation, knowledge transfer, strategies and challenges, which later helped us in data analysis. The duration of these interviews averaged 60 minutes and the interview sessions were tape recorded. Among the seventeen semi-structured interviews, nine were conducted through Skype and eight were face-to-face, due to the distance.

### C. Data Analysis

This research adopted thematic qualitative data analysis technique [14] to analyze the transcribed raw data and was applied by the first author. Thematic analysis is used to identify, analyze and report patterns or themes within data. In thematic analysis a theme captures data with relation to research question and represents them in a pattern within the data set [14]. The analysis was performed through a process which maintain six phases as suggested by Braun and Clarke [14]: familiarization with data, generating initial codes, searching for themes among code, reviewing themes, defining and naming themes, and producing the final report.

In the *first* stage, the first author transcribed all the collected interview data into written form in order to conduct a thematic analysis. It helped us to identify possible themes and patterns, and develop potential codes [15]. *Second* phase started with initial codes from the extracted data. In this phase, the open coding technique was applied to relate coded data with research theme and research questions. In the *third* phase, we the codes and grouped them into broader categories or themes [14]. Later, mind mapping tools were used to represent them into theme-piles. This stage gave us a sense of the significance of individual themes. Stage *four* is reviewing themes. In this stage we identified irrelevant (not enough or diverse) data with related to different themes. After refining all themes we identified “essence” of each theme and different aspects of

the data each theme captures in stage *five*. At the end, in stage *six*, we provided extracted data with relation of the research question and presented in support of results and discussion sections.

### D. Validity Threats

To handle validity threats it is important to identify all possible factors that might affect the accuracy or dependability of the results. The validity of our results might have been affected by a number of threats. First of all, internal validity could be affected by the sampling of the interviewees. We were not able to interview all team members in all locations. However, we have tried to mitigate this threat by interviewing several team members in each of the projects thus applying triangulation techniques to the data collected from different interviewees. Interpretation validity was maintained by recording the interviews and performing the data analysis using the transcribed records, which were to the interviewees for additional confirmation. There exist threats to external validity of our results too. We encountered major challenges in relation to the willingness and availability of potential interviewees and thus only seventeen interviews in six companies were conducted. Notably, the choice of the companies followed convenience sampling. This means that we cannot generalize our results to all distributed agile projects. However, we have tried to address these threats by selecting different type of projects, which are also geographically spread, and found similar findings.

## IV. KNOWLEDGE MANAGEMENT IN SEVEN DISTRIBUTED AGILE PROJECTS

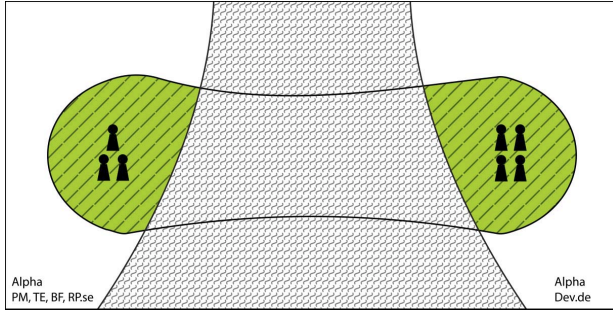
This section describes the empirical cases anonymously denoted as alpha, beta, gamma, delta, epsilon, zeta, and eta. When discussing our findings we distinguished between three types of teams in distributed projects: distributed teams, dispersed teams, and hybrid teams [2]. Based on the team size, we have also divided the cases into three categories: small, medium, and large teams. Tables III-V provide overviews of the studied projects.

### A. Small Teams

**TABLE III:** Small teams

| Projects     | Team Distribution | Development Method(s)      |
|--------------|-------------------|----------------------------|
| <b>Alpha</b> | Sweden-Germany    | Scrum and Kanban           |
| <b>Beta</b>  | Norway-Bangladesh | Feature-driven development |

1) *Alpha*: Alpha is a software company based in Stockholm, Sweden, which has offshore sites in different countries. The project studied is a collaboration of team members from Sweden and Germany working in a dispersed team (see Figure 1; green color indicates that the interviews were conducted with members of the both teams). The Swedish site is responsible for the overall project management, testing, feature development, bug fixing and release planning. The German site is responsible for core development and testing. The interviewees were from the Stockholm project management team and the German development team.

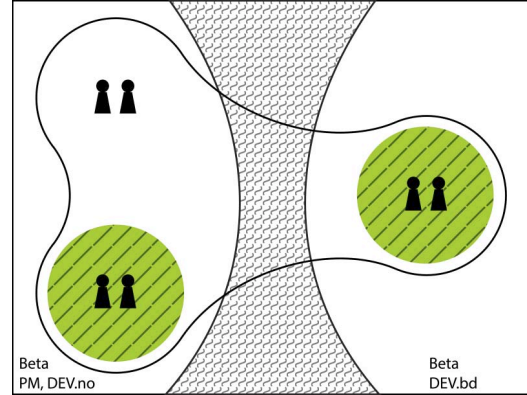


**Fig. 1:** Team distribution in Alpha project

As a development methodology, Alpha practices agile, which is a combination of both Scrum and Kanban. Stockholm team begins the day with a startup meeting. After the meeting, all task-related information is kept on the Scrum board. Redmine is strictly maintained by the team members for documentation and to share knowledge among both local and distributed team members. There are no scrum and Kanban boards at the German site. All the information from the Scrum and Kanban boards is codified into Redmine by the Swedish team in order to share and exchange this knowledge among the German team members. Skype is used as the major communication tool among the remote team members. Every week they have a weekly workshop with the local team, lasting 1-2 hours; this helps team members to share their thoughts with each other. Though the German team does not have a Scrum board, the team accesses the information through Redmine.

2) *Beta*: Beta, is a software company based in Oslo, Norway. The company has three sites in two countries working as one team on the same project. There are two sites in Oslo. One site (green color) is responsible for project management and some parts of development, and other site is responsible for testing. The Bangladeshi site is responsible for core

development. Team structure- *Dispersed Team* (see in Figure 2).



**Fig. 2:** Team distribution in Beta project

As a development methodology the Beta team practices agile, more specifically feature-driven development (FDD). Redmine, GitHub and Skype are used to share features and scheduling, code and screen respectively in the local team. Wiki tools are used for documentation. The team does not have any physical boards in their respective sites, team members share all information through Redmine and Wikis for both local and remote use.

The project manager said “*I am not satisfied with the current tools, It's tough to describe designs to new team members. Visual aids are helpful during discussions*”.

### B. Medium Teams

**TABLE IV:** Medium teams

| Projects       | Team Distribution | Development Method(s) |
|----------------|-------------------|-----------------------|
| <b>Gamma</b>   | USA-Bangladesh    | Scrum                 |
| <b>Delta</b>   | Sweden-Bangladesh | Agile; Waterfall      |
| <b>Epsilon</b> | Latvia-Ukraine    | Scrum, XP, and Kanban |

1) *Gamma*: The core business of this company is design and development. Gamma project used a closely observed software development life cycle (SDLC) and relied on lightweight, productive and client-friendly processes. At the beginning they started with traditional waterfall SDLC model, but it is no longer used due to today's fast-paced industry where change is the only constant. Now, the Gamma team is using an interactive agile model; continuously iterates from initial requirements until initial launch and onward until the final launch. Though design and development are the core business in the company, they also provide testing and QA, integration and IT, tech support, maintenance and refactoring services. A small team with the responsibility for project management, business analysis and architecture is situated in San Jose, CA, USA and a large engineering, IT and technical support team is situated in Dhaka, Bangladesh. According to our classification this means that the project employs two *distributed teams* (see in Figure 3).

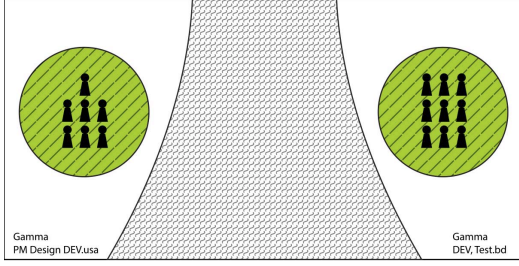


Fig. 3: Team distribution in Gamma project

In one of the current projects with the Dhaka team, the USA team is responsible for the core design and decision-making. As a development methodology they practice agile, more specifically, they use Scrum. The Dhaka team is responsible for development and testing. So, they start the day with a short meeting with the USA team. The USA team holds the main Scrum board and the Dhaka team holds a sub-scrum board (for a specific sprint). To share central knowledge with the local team they maintain a knowledge management tool. In the distributed environment they maintain a wikis for documentation sharing, GitHub for code sharing and Skype for communication. To share the knowledge from the physical scrum board with the remote team, the USA team uploads pictures every day onto the wikis.

The scope of the knowledge sharing is described by the interviewee: “It’s ok for a small team, but an electronic white board would be helpful to share knowledge among remote team members.”

2) *Delta*: Delta, provides another example of dispersed teams: these are distributed between Sweden and Bangladesh. The Swedish team is responsible for both specification and design phase; on the other site, the Bangladeshi team is responsible for core development.

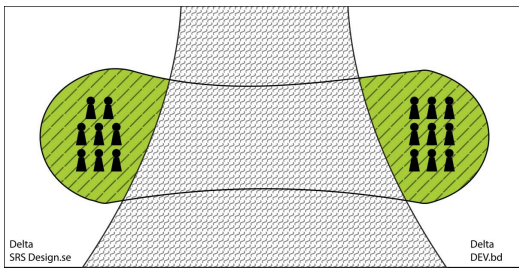


Fig. 4: Team distribution in Delta project

As a development methodology they have adopted agile but not specifically any practices like Scrum or Kanban. Their motto is: *Whatever works!* For task-related knowledge sharing they maintain GreenHopper Atlassian JIRA and they do not have any physical boards in their offices. Confluence is used for knowledge management. If someone faces any problems

in the local team they ask the *Sage*<sup>1</sup> directly, in order to get an immediate answer rather than searching. But for the remote team task-related knowledge is kept in the sprint management tool and for the domain-related knowledge they maintain a common chat room to resolve specific problems.

The *software architect* of this company said “*Visual aids are helpful if diagrams or other issues are hard to describe*”.

3) *Epsilon*: Epsilon is an based IT company providing banking and finance solutions based in Riga, Latvia. It is one of the fastest growing IT service providers in Latvia. Epsilon specializes in providing IT consulting, as well as IT systems development, integration and software maintenance services.

One of the current projects is a collaboration between Latvia and Ukraine. The Latvian team is responsible for project management, requirement analysis, design and verification; and the Ukraine team is responsible for core development. The team uses agile, which is a mixture of Scrum, XP and Kanban. It is not used in a strict way: it is tailored and flexible.

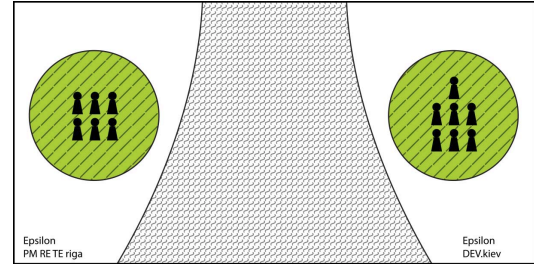


Fig. 5: Team distribution in Epsilon project

The Latvian team starts the day with a short meeting. All project and domain related information is kept in the wikis and JIRA. The main challenges faced by the Epsilon team, concerning the sharing of knowledge among the distributed team, are distance and visualization. So, to mitigate those challenges they maintain a common platform (the same work environments), a common chat room and occasionally visit the remote site.

One of the interviewees said “*It is good to have all the information on the electronic board for distributed teams*”.

### C. Large Teams

TABLE V: Large teams

| Projects    | Team Distribution | Development Method(s) |
|-------------|-------------------|-----------------------|
| <b>Zeta</b> | Sweden-China      | Scrum & Kanban        |
| <b>Eta</b>  | Sweden-India      | Scrum & Kanban        |

1) *Zeta*: Zeta, is one of the leading companies in the telecoms industry and was founded in Sweden. One of the current projects is a collaboration between *Sweden* and *China*, where two distributed teams are working on one project.

<sup>1</sup> Sage is usually an older person of experience. Can give advice or guidance to the younger team members.

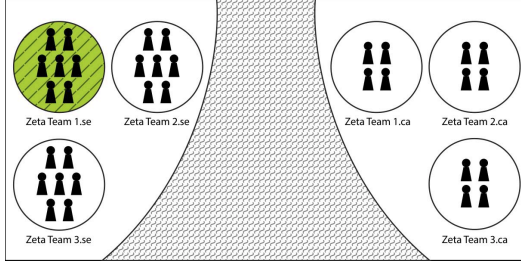


Fig. 6: Team distribution in Zeta project

As a development methodology both teams practice *agile*, more specifically, *Scrum* and *Kanban*. To share knowledge with the local team they maintain a wikis, pair programming, technical presentation and a technical discussion forum. But to share knowledge with the remote team they perform a daily conference, sometimes visiting the remote site, and a technical forum, called as a *knowledge hub*.

The Line Manager said “We are planning to buy a digital scrum board to increase the communication bandwidth, which will help distributed teams to get information at an early stage”.

2) *Eta*: Eta, is also one of the leading companies in the telecoms industry in Sweden. One of their current projects is a collaboration between six distributed teams and every team has two remote team members in *India*. The team structure is *hybrid* and one of the *Stockholm* teams is responsible for the project management.

As a development methodology the team uses *agile* which is a mix of both *Scrum* and *Kanban*, but not in a strict manner. To share knowledge among distributed dispersed teams both formal and informal processes are followed by the team members. Most of the time workshops and seminars are arranged on different topics such as technologies, tools and design methods to create local knowledge. Also, team meetings and on-demand meetings are maintained for knowledge sharing. To share knowledge across the sites the teams use electronic media (such as electronic boards, wikis, forums, documentation, etc.), online conferences, online meetings, and site visits.

One interviewee mentioned that “It seems people sitting near to each other find it easier to communicate and apparently access to knowledge and sharing of knowledge increase. However, electronic mediums are functioning well enough to share knowledge. Electronic boards are in place containing the tasks to perform list and latest information. Along with that, necessary technical and business information is regularly updated on the Wikis, the internal web, etc. Online forums are also used to discuss”.

The interviewee also suggested that “Knowledge sharing, within the team and between the teams is a necessary function to be able to perform effectively. This effectiveness includes timeliness, accuracy in design and development, and following the right technical and business processes”.

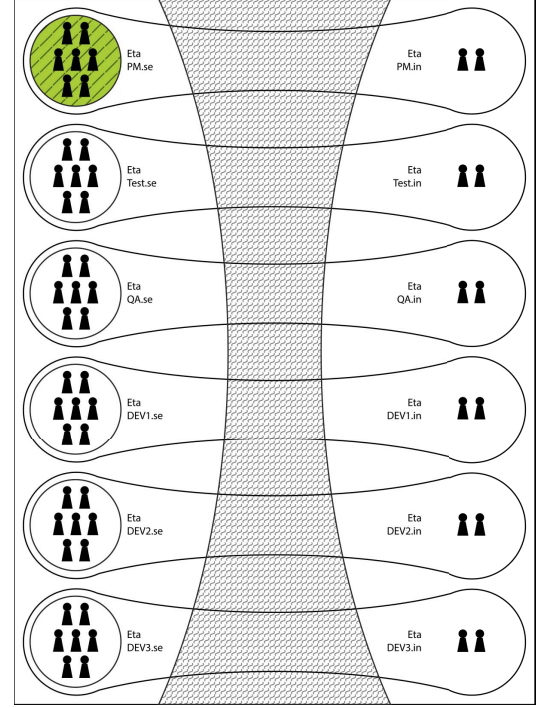


Fig. 7: Team distribution in Eta project

## V. DISCUSSION

### A. Knowledge Management Strategies in Practice

We were interested to see whether all knowledge management schools are in practice. We used Earl’s [6] framework to select types of strategies called “knowledge management schools” and checked whether we find practices that can be mapped to these strategies in the different projects (See Table VI). We were also interested to see whether these strategies are applied locally and/or globally (denoted by letters L and G in the table).

Based on the evidence collected, we found that companies used a variety of practices that correspond to many different schools. In summary, we found systems, engineering, and organizational schools to be highly practiced in the studied distributed agile projects to manage knowledge both locally and globally. Cartographic school practices were used only in three cases but supported team members both locally and globally. But spatial practices were found primarily locally. Both *commercial* and *strategic* schools are focused on a business perspective (*patent, copyright, trademark, know-how* and *intellectual assets* [6]) and there was no evidence found in the studied cases that indicates those schools are in practice. For that reason we excluded those schools from further analysis.

1) *Systems school*: The systems school was in practice in all studied cases to manage knowledge both locally and globally. Though documented knowledge is not always easy to find and extract, we found that the software engineers in the case companies depended on the codified knowledge because the use of knowledge databases and repositories considerably

**TABLE VI:** Knowledge sharing strategies in practice

| <i>KM Schools</i> | $\alpha$ | $\beta$ | $\gamma$ | $\delta$ | $\varepsilon$ | $\zeta$ | $\eta$ |
|-------------------|----------|---------|----------|----------|---------------|---------|--------|
| System            | L,G      | L,G     | L,G      | L,G      | L,G           | L,G     | L,G    |
| Cartographic      | —        | —       | —        | —        | L,G           | L,G     | L,G    |
| Engineering       | L,G      | L,G     | L,G      | L,G      | L,G           | L,G     | L,G    |
| Organizational    | L,G      | L       | L,G      | L,G      | L,G           | L,G     | L,G    |
| Spatial           | L        | —       | L        | L        | —             | L       | L,G    |

\*In Table VI, L indicates Locally, G— Globally and “—” not in practice

alleviated the knowledge sharing activities across remote locations. From the studied cases, we learned that, these knowledge repositories usually contained domain specific knowledge. Our findings confirm the popularity of knowledge management practices under the systems school, which is also the most researched school [7].

2) *Cartographic school*: In collocation, it seems easier to find a knowledgeable or experienced person because they know each other well. In globally distributed projects “who knows what” and “what is where” are important issues for effective knowledge exchange. We have found that the *cartographic* school is practiced by some projects ( $\delta, \zeta, \eta$ ) to exchange knowledge both locally and globally. We have also found evidence about dedicated knowledge brokers from one case where team introduce the idea of knowledge brokers<sup>2</sup> which helps team members to consult with knowledgeable and experienced software engineers.

3) *Engineering school*: This school of knowledge management focuses on business process re-engineering [7] and knowledge flows in organizations. More specifically, this school focuses on formal routines, mapping of knowledge flows, project reviews, and social interactions. This school supports explicit knowledge sharing and thus the corresponding knowledge management practices in distributed projects are not affected by temporal distance. Since coordination is one the major challenges in distributed projects, the engineering school is of high importance as it focuses on knowledge coordination through knowledge flows and shared databases. We found that processes of using tools (*i.e. the installation manual for GitHub or SVN with Eclipse*), coding standards, testing and review processes are all documented in repositories to be used by team members in all locations. Finally, similarly to the systems school, we found that, all studied cases follow knowledge management practices that can be mapped to the engineering school, both locally and globally.

4) *Organizational school*: The philosophy of the *organizational* school is related to creation of collaboration networks and communities that facilitate sharing and pooling knowledge resources. This school of knowledge management focuses on organizational structure, which are often referred to as “knowledge communities” [7]. This is a networking approach for people to communicate and share knowledge. As we know, successful development and deployment depends on the shared understanding between stakeholders and software engineers.

<sup>2</sup>Knowledge brokers are knowledgeable and experienced software engineers who will communicate with other developers, provide them with information or listen to them.

This is why practices that fall under this strategy are very important. We have found that team members share knowledge through wikis and forums. These practices also help team members to develop a shared understanding of the processes and procedures.

5) *Spatial school*: The intention of the spatial school is to encourage socialization (tacit to tacit) as a means of knowledge exchange [16]. The spatial school is more concerned with the development and utilization of the social capital which develops from people interactions, formal or informal, repeatedly over time [6]. The spatial school focuses on designing office spaces to promote knowledge sharing [5]. Organizations use different office settings to promote communication between people such as conference rooms, dining rooms or places for refreshment and activities that all facilitate knowledge sharing. So knowledge is created by learning during the discussion in front of coffee machines, in the meetings, during training and teamwork [17–20]. Among the studied cases, five projects reported practices that can be mapped to the spatial school. As expected, the majority of these practices were used locally. These were discussions in front of agile Scrum boards and charts, as well as the use of space near coffee machines. Only one case (*hybrid team, \eta*) reported application of spatial practices to manage knowledge both locally and globally. The hybrid team used visual boards in local discussions and during the video-conferences with remote team members.

## B. Lessons Learned

There is a common belief that agile development neglects documentation, supported by empirical evidence that emphasizes the lack of documentation reported by practitioners working in agile projects [21]. In fact, agile software development can be described as collaboration-centered, in which knowledge management is achieved primarily through teamwork and informal knowledge exchange. Surprisingly, we found from our results that companies in all studied cases are concerned about knowledge codification. Teams members document their knowledge using Wikis, JIRA, Confluence etc. Even the knowledge that is shared informally between the collocated team members during technical presentations and discussions, is later codified to be shared with the remote team members. Thus we suggest that

- traditional knowledge management strategies applied in agile software development require adjustments when team members are dispersed or distributed, and
- knowledge sharing in distributed projects require effective knowledge codification practices.

According to Polanyi [22], “*individuals know more than they can say*”. Polanyi classified human knowledge into two categories –explicit knowledge and tacit knowledge. Explicit knowledge is easily characterized in textual or symbolic forms, written down and codified. In contrast, tacit knowledge is very difficult to describe or express, and it is transferred through demonstration. Tacit knowledge has an important cognitive dimension, which consists of mental models, beliefs and perspectives [17,23,24]. So it often cannot be easily characterized

by clear expressive language. Codification of knowledge is the conversion of tacit knowledge into explicit knowledge in a written, verbal or visual format. Although codification helps to convert tacit knowledge into explicit, managing tacit knowledge, which dominates agile development, is a challenging task. The transfer of tacit knowledge into explicit, knowledge, which is also referred to as externalization [25], often causes knowledge loss, since tacit knowledge cannot be interpreted fully even by an expert [26,27]. This type of knowledge is more deeply placed in action and is hard to express in words [28]. This also means that knowledge codification alone might be insufficient to effectively share the knowledge. We thus suggest that

- effective knowledge requires careful planning and selection of keen experts that understand the essence of the tacit knowledge in order to increase the degree of explicitness of knowledge, and
- remote team members shall find a way to socialize, interact virtually and perhaps even simulate a shared space for creating and exchanging tacit knowledge.

## VI. CONCLUSIONS

Software development is a knowledge intensive job and it highly depends on the team members' (developers, software architect, QA etc.) knowledge and experience. Distributed projects introduce new challenges to knowledge management, so, practices that are used in agile teams become difficult to put into action in distributed development. However, inadequate knowledge management may doom distributed agile projects to failure. One of the biggest challenges in the organizations is to find the right knowledge at the right time due to knowledge fragmentation, overload and de-contextualization. So, it is important for the globally distributed organization to establish and maintain an effective knowledge management strategy for both tacit and explicit knowledge creation and sharing.

In response to the research question (*Which knowledge management strategies are used by distributed agile teams and how?*) we conclude that knowledge sharing across remote locations heavily relies on knowledge codification, i.e. technocratic KM strategies, even when the same knowledge is shared tacitly within the same location, i.e. through behavioral KM strategies. Our results suggest that distributed agile projects teams in our study practiced a number of different strategies to perform knowledge creation and sharing. We found practices that were mapped to systems, engineering, organizational and cartographic schools as categorized by Earl [6] both locally and globally. We also found that practices that can be mapped to the spatial school are primarily used locally, perhaps because tools that could enable the use of such practices virtually are still very expensive. Similarly practices that help to achieve awareness of who knows what and what is where that are associated with the cartographic school of knowledge management were not widely applied among the studied cases.

**Future Works:** While this research focused on how team members contribute to knowledge creation and sharing in

distributed agile projects, it would be interesting to further evaluate the effectiveness of the found practices and seek the gaps in the shared understanding or awareness between remote team members. Another potential research direction is related to the virtual simulation of the collocated knowledge management practices as applied in agile. In particular, it would be interesting to see whether socialization and tacit to tacit knowledge sharing through virtual technology could diminish the necessity to codify the knowledge in distributed projects.

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